

Online Bookstore Review and Catalog

An E-Commerce Platform for Buying and Selling Books Online

GROUP NUMBER: G2

1. Proposed Project Description

1.1 Overview

We propose to build **Online Bookstore Review and Catalog**, a web-based e-commerce platform for buying and selling books online. This is a brand-new project rather than an extension of one of the sample software projects, though it draws on the familiar structure of an online retail system.

The platform combines three concerns that are usually spread across separate sites: a **searchable catalog** of books with rich metadata, a **community review and rating layer** that helps buyers judge a book before purchasing, and a **two-sided marketplace** where individual sellers can list their own copies — new or used — alongside the main inventory. A buyer can therefore discover a book, read genuine reviews written by people who actually purchased it, compare listings from different sellers, and complete the purchase without leaving the system.

The problem we are addressing is a practical one. Existing large marketplaces bury book-specific information under generic product pages, while dedicated review sites offer no purchase path at all. Readers end up switching between two or three sites to make a single buying decision. Small and independent sellers, meanwhile, face high listing fees and heavy onboarding on large platforms. Our system targets the gap between these two.

1.2 Customer and Target Users

We do not have a committed external customer at this stage. The project is being developed against a realistic requirements specification that we have authored and reviewed as a team, and we have informally validated the idea with fellow students who buy and resell second-hand textbooks each semester — a group that represents our most immediate potential user base.

The system is designed for three classes of users:

- **Buyers** — students, casual readers and collectors who want to browse or search a catalog, read reviews, and purchase books. This is the largest user group and the primary focus of the interface.
- **Sellers** — individuals or small bookshops who want to list books for sale, manage inventory and stock levels, and fulfil incoming orders without needing technical expertise.
- **Administrators** — the platform operators, responsible for moderating listings and reviews, managing users and sellers, maintaining category structures, and resolving disputes.

A typical user is assumed to be comfortable with common online shopping interfaces but not technically expert, and may access the system from either a desktop browser or a mobile device.

1.3 Technology Stack

The system will be implemented as a **MERN stack** application:

- **MongoDB** — document database for books, users, listings, orders and reviews
- **Express.js** — REST API layer
- **React** — single-page front end, responsive across screen sizes
- **Node.js** — server runtime

We evaluated Python/Django with Django REST Framework as an alternative. MERN was selected because the flexible document model suits the varying metadata across book listings (condition, edition, seller-specific fields), and because a single language across client and server reduces context switching for a four-person team working in parallel. Version control and code review will be managed through GitHub, using feature branches and pull requests.

1.4 Functional Requirements

At the conclusion of this project, a user of the system will be able to:

1. **Register an account** using their personal details. *(FR01)*
2. **Log in and log out** securely as a registered user. *(FR02)*
3. **View and update their profile** and delivery address. *(FR03)*
4. **Browse the book catalog**, seeing title, author, price, category, ISBN, description and availability. *(FR04)*
5. **Search for books** by title, author, ISBN or free-text keyword. *(FR05)*
6. **Filter and sort books** by category, price, rating and availability. *(FR06)*
7. **View a book's detail page**, including full information, aggregate rating and individual reviews. *(FR07)*
8. **Manage a shopping cart** — add books, remove books and change quantities. *(FR08)*
9. **Check out an order** by selecting a delivery address and reviewing the order before placing it. *(FR09)*
10. **Pay for an order** using the available payment methods. *(FR10)*
11. **Place orders and track them**, viewing complete order history and current status. *(FR11)*
12. **Rate and review books** they have purchased. *(FR12)*
13. **List books for sale** as a seller, specifying price, quantity, condition and other details. *(FR13)*
14. **Add, edit and remove their own listings** as a seller. *(FR14)*
15. **Administer the platform** — manage users, books, sellers, orders, categories and reviews. *(FR15)*
16. **Receive notifications** about orders, payments and order status changes. *(FR16)*

2. Plan of Work and Product Ownership

2.1 Short-Term Plan of Work

The following items are planned for the next few weeks.

Week	Planned Work
Week 1	Finalise requirements; freeze the FR/NFR list; agree on the MongoDB schema for users, books, listings, orders and reviews; set up the GitHub repository, branching strategy and issue board.
Week 2	Scaffold the project: Express server, MongoDB connection, React application shell, shared routing and component conventions. Implement authentication (registration, login, JWT-based sessions). Seed the database with sample book data.
Week 3	Build the catalog: book listing pages, book detail page, search endpoint and filter/sort logic. Begin the seller listing module.
Week 4	Build the transaction path: shopping cart, checkout flow, address selection and order creation. Complete seller listing management.
Week 5	Integrate payment (sandbox gateway), order history and status tracking, and notifications. Implement reviews and ratings tied to verified purchases.
Week 6	Admin module, unit and integration testing, performance tuning against NFR targets, responsive and accessibility pass, documentation and final demonstration.

Testing is not deferred to the end: each member writes unit tests for their own modules as they are built, and the team runs an integration review at the close of each week.

2.2 Product Ownership

Ownership is assigned so that each member holds an end-to-end vertical slice of the system — database schema, API endpoints and front-end views for their features — together with one qualitative property of the overall system.

Selvaganesh Arunmozhi — PES2UG24CS451

Role: Accounts, Access Control and Administration

Functional features owned:

- FR01 — User Registration
- FR02 — User Login
- FR03 — User Profile
- FR15 — Admin Management

Qualitative property: Security and privacy of the system (NFR02, NFR11). Will implement password hashing with bcrypt, JWT-based session handling and role-based access control across all protected endpoints, and will verify that no user's personal information is retrievable by an unauthorised account through any API route.

Shreya Bonal — PES2UG24CS485

Role: Catalog, Search and Discovery

Functional features owned:

- FR04 — Book Catalog
- FR05 — Search Books
- FR06 — Filter and Sort
- FR07 — View Book Details

Qualitative property: System performance (NFR01). Will tune database indexing and query structure and add pagination and lazy loading, with the target that catalog and search result pages return within 3 seconds — and search queries specifically within 2 seconds — on a seeded database of at least 5,000 books.

Sharanabasava Shivappa — PES2UG24CS457

Role: Cart, Checkout, Payments and Orders

Functional features owned:

- FR08 — Shopping Cart
- FR09 — Checkout
- FR10 — Payment
- FR11 — Order Management
- FR16 — Notifications

Qualitative property: Reliability and data integrity of transactions (NFR04, NFR09). Will ensure that order placement, payment confirmation and stock decrement are handled atomically, so that no order is recorded without a corresponding inventory update and a failed or interrupted payment leaves no partial order in the database.

Satvik Das — PES2UG24CS448

Role: Reviews, Ratings and the Seller Marketplace

Functional features owned:

- FR12 — Reviews and Ratings
- FR13 — Sell Books
- FR14 — Manage Listings

Qualitative property: Usability and accessibility (NFR03, NFR08, NFR12). Will design and evaluate the seller listing and review submission interfaces so that a new seller can publish a complete listing in under five steps, and will ensure both flows are keyboard-navigable, readable at standard contrast ratios and usable at mobile screen widths.

2.3 Shared Responsibilities

The following are owned jointly by the full team and are not counted as functional features:

- Database schema design and review
- Common React component library and styling conventions
- Code review on all pull requests, with a minimum of one reviewer other than the author
- Integration testing and bug triage
- Project documentation, SRS maintenance and the final report
- Deployment configuration and demonstration preparation

3. Non-Functional Requirements Summary

The system will be evaluated against the following quality attributes, each mapped to an owner in Section 3.2 where applicable.

ID	Property	Target
NFR01	Performance	Major pages load within 3 seconds under normal conditions
NFR02	Security	User accounts, passwords and sensitive information are protected
NFR03	Usability	Simple, intuitive and user-friendly interface
NFR04	Reliability	Orders and transactions processed accurately without data loss
NFR05	Availability	Minimal downtime for users
NFR06	Scalability	Supports increasing numbers of users, books, sellers and orders
NFR07	Maintainability	Modular, well-documented code
NFR08	Compatibility	Works across major browsers and screen sizes
NFR09	Backup and Recovery	Regular backups with recovery after failure
NFR10	Privacy	No unauthorised access to users' personal information
NFR11	Accessibility	Clear navigation, readable text, accessible interface elements